IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

This project employed a structured data science methodology, initiating with data collection through the SpaceX REST API and web scraping to aggregate historical Falcon 9 launch records. Subsequent data wrangling involved cleaning missing values and defining a binary classification target for landing outcomes, followed by Exploratory Data Analysis (EDA) using SQL for querying, Seaborn for visualization, and Folium for geospatial mapping to identify patterns in launch sites and orbits. Feature engineering techniques, particularly One-Hot Encoding, prepared the categorical data for machine learning, where four distinct models, Logistic Regression, Support Vector Machine (SVM), Decision Tree, and K-Nearest Neighbors (KNN), were trained, standardized, and optimized using GridSearchCV with 10-fold cross-validation. The results demonstrated a convergence in model performance, with all classifiers achieving a test accuracy of approximately 83.33% (identifying the Decision Tree as the top performer on training data but equal on test data), indicating that while the specific algorithm had minimal impact on this dataset, the underlying features such as payload mass, orbit type, and launch site were robust predictors of first-stage landing success.

Introduction

Project Background and Context

SpaceX has revolutionized the aerospace industry by making space travel more affordable through **reusability**. unlike traditional providers who discard costly rocket stages into the ocean, SpaceX has developed the technology to land and reuse the first stage of its **Falcon 9** rocket. This technological advantage allows SpaceX to offer launches at a significantly lower price point, approximately **$62 million** per launch, compared to competitors like United Launch Alliance (ULA), whose expendable launches often cost upwards of **$165 million**.

The Problem

The primary objective of this project is to **predict whether the Falcon 9 first stage will land successfully**. By determining the probability of a successful landing using historical data (such as payload mass, orbit type, and launch site), we can accurately estimate the cost of a launch. This predictive capability is crucial for making competitive bids against other space launch providers; if we know the rocket will land, the operational cost is lower, allowing for a more aggressive bid.

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
  - The SpaceX launch data is typically collected by combining direct calls to the **SpaceX REST API web scraping** of public sources like Wikipedia (to extract historical tables)
- Perform data wrangling
  - The data processing involved **cleaning** the dataset to handle missing values (like payload mass) and **classifying** the landing outcomes into a binary target variable (1 for success, 0 for failure).
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
  - How to build, tune, evaluate classification models

Data Collection

1. SpaceX REST API

This is the most direct and reliable source for up-to-date and structured data.

- **Method:** Making HTTP requests (specifically GET requests) to the official SpaceX API endpoints (usually version 4, e.g., api.spacexdata.com/v4/).
- **Data Format:** The API returns data in **JSON** format.
- **Process:** Developers fetch the JSON data, parse it, and then **normalize** the sometimes complex, nested structure into a flat, tabular format (like a CSV or a database table) that is suitable for analysis. This method captures comprehensive details about each launch, including flight number, payload mass, and launch outcome.

2. Web Scraping (Wikipedia)

For historical data, especially launch results and booster information, web scraping the Falcon 9 Wikipedia page is a common supplemental method.

- **Method:** Using libraries like **BeautifulSoup** in Python to extract the large HTML table data from the relevant Wikipedia page (e.g., "List of Falcon 9 and Falcon Heavy launches").
- **Data Format:** The data is initially unstructured HTML table content.
- **Process:** The content is extracted, cleaned (removing references, converting units), and then transformed into a structured DataFrame. This is particularly useful for verifying booster serial numbers and specific landing outcomes before the modern API was fully comprehensive.

By combining these two methods, a complete and detailed dataset on Falcon 9 launches, landing attempts, and mission outcomes is created, which is then loaded into a tool or database (like the one we've been simulating) for analysis and visualization.

Data Collection – SpaceX API

- Present your data collection with SpaceX REST calls using key phrases and flowcharts

- Add the GitHub URL of the completed SpaceX API calls notebook (must include completed code cell and outcome cell), as an external reference and peer-review purpose

Place your flowchart of SpaceX API calls here

Data Collection - Scraping

- Present your web scraping process using key phrases and flowcharts

- Add the GitHub URL of the completed web scraping notebook, as an external reference and peer-review purpose

Place your flowchart of web scraping here

Data Wrangling

- **Data Filtering:**
  - We filtered the dataset to include **only Falcon 9** launches, removing data related to Falcon 1 and Falcon Heavy, as the project focus is specifically on Falcon 9 first-stage landings.
- **Handling Missing Values (Imputation):**
  - **Payload Mass:** Some launches had missing payload mass data. We calculated the **mean (average) payload mass** of the known values and used it to fill in the missing entries.
  - **Landing Pads:** Missing landing pad values were handled to ensure the model didn't fail on null inputs.
- **Class Classification (Target Variable):**
  - The original Landing_Outcome column contained various string descriptions (e.g., "Success (drone ship)", "Failure (in flight)", "No attempt").
  - We converted this into a **binary classification** target variable named Class:
    - **1** = Successful landing.
    - **0** = Unsuccessful landing (failure or no attempt).
- **Feature Engineering (One-Hot Encoding):**
  - Categorical variables such as Orbit, LaunchSite, LandingPad, and Serial (booster version) cannot be understood directly by machine learning models.
  - We applied **One-Hot Encoding** (using pd.get_dummies) to convert these text labels into binary numeric columns (e.g., Orbit_LEO, Orbit_GTO).
- You need to present your data wrangling process using key phrases and flowcharts

EDA with Data Visualization

- Summarize what charts were plotted and why you used those charts
- Add the GitHub URL of your completed EDA with data visualization notebook, as an external reference and peer-review purpose

EDA with SQL

- **Identified the First Ground Pad Landing:** Used the MIN(Date) function with a specific WHERE clause to pinpoint the exact date of the first successful landing on a ground pad (December 21, 2015).
- **Filtered Boosters by Performance and Payload:** Queried the database to list specific booster versions that achieved a successful drone ship landing while carrying a payload between **4,000 kg and 6,000 kg**.
- **Analyzed Mission Outcomes:** Used GROUP BY and COUNT() to aggregate the data, providing a breakdown of total successful versus failed missions.
- **Identified Heavy-Lift Boosters:** Employed a **subquery** to first calculate the maximum payload mass (MAX()) in the entire dataset, and then listed all booster versions associated with that maximum weight.

Build an Interactive Map with Folium

- Summarize what map objects such as markers, circles, lines, etc. you created and added to a folium map
- Explain why you added those objects
- Add the GitHub URL of your completed interactive map with Folium map, as an external reference and peer-review purpose

Build a Dashboard with Plotly Dash

- Summarize what plots/graphs and interactions you have added to a dashboard
- Explain why you added those plots and interactions
- Add the GitHub URL of your completed Plotly Dash lab, as an external reference and peer-review purpose

Predictive Analysis (Classification)

- Summarize how you built, evaluated, improved, and found the best performing classification model
- You need present your model development process using key phrases and flowchart
- Add the GitHub URL of your completed predictive analysis lab, as an external reference and peer-review purpose

Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

- Show a scatter plot of Flight Number vs. Launch Site

- Show the screenshot of the scatter plot with explanations

Payload vs. Launch Site

- Show a scatter plot of Payload vs. Launch Site

- Show the screenshot of the scatter plot with explanations

Success Rate vs. Orbit Type

- Show a bar chart for the success rate of each orbit type

- Show the screenshot of the scatter plot with explanations

Flight Number vs. Orbit Type

- Show a scatter point of Flight number vs. Orbit type

- Show the screenshot of the scatter plot with explanations

Payload vs. Orbit Type

- Show a scatter point of payload vs. orbit type

- Show the screenshot of the scatter plot with explanations

Launch Success Yearly Trend

- Show a line chart of yearly average success rate
- Show the screenshot of the scatter plot with explanations

All Launch Site Names

- Find the names of the unique launch sites
- Present your query result with a short explanation here

Launch Site Names Begin with 'CCA'

- Find 5 records where launch sites begin with `CCA`
- Present your query result with a short explanation here

Total Payload Mass

- Calculate the total payload carried by boosters from NASA
- Present your query result with a short explanation here

Average Payload Mass by F9 v1.1

- Calculate the average payload mass carried by booster version F9 v1.1
- Present your query result with a short explanation here

First Successful Ground Landing Date

- Find the dates of the first successful landing outcome on ground pad
- Present your query result with a short explanation here

Successful Drone Ship Landing with Payload between 4000 and 6000

- List the names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000

- Present your query result with a short explanation here

Total Number of Successful and Failure Mission Outcomes

- Calculate the total number of successful and failure mission outcomes
- Present your query result with a short explanation here

Boosters Carried Maximum Payload

- List the names of the booster which have carried the maximum payload mass
- Present your query result with a short explanation here

2015 Launch Records

- List the failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015

- Present your query result with a short explanation here

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order

- Present your query result with a short explanation here

Section 3

Launch Sites Proximities Analysis

<Folium Map Screenshot 1>

- Replace <Folium map screenshot 1> title with an appropriate title

- Explore the generated folium map and make a proper screenshot to include all launch sites' location markers on a global map

- Explain the important elements and findings on the screenshot

<Folium Map Screenshot 2>

- Replace <Folium map screenshot 2> title with an appropriate title

- Explore the folium map and make a proper screenshot to show the color-labeled launch outcomes on the map

- Explain the important elements and findings on the screenshot

<Folium Map Screenshot 3>

- Replace <Folium map screenshot 3> title with an appropriate title
- Explore the generated folium map and show the screenshot of a selected launch site to its proximities such as railway, highway, coastline, with distance calculated and displayed
- Explain the important elements and findings on the screenshot

Section 4

Build a Dashboard with Plotly Dash

<Dashboard Screenshot 1>

- Replace <Dashboard screenshot 1> title with an appropriate title
- Show the screenshot of launch success count for all sites, in a piechart
- Explain the important elements and findings on the screenshot

<Dashboard Screenshot 2>

- Replace <Dashboard screenshot 2> title with an appropriate title

- Show the screenshot of the piechart for the launch site with highest launch success ratio

- Explain the important elements and findings on the screenshot

<Dashboard Screenshot 3>

- Replace <Dashboard screenshot 3> title with an appropriate title
- Show screenshots of Payload vs. Launch Outcome scatter plot for all sites, with different payload selected in the range slider
- Explain the important elements and findings on the screenshot, such as which payload range or booster version have the largest success rate, etc.

Section 5

Predictive Analysis (Classification)

Classification Accuracy

- Visualize the built model accuracy for all built classification models, in a bar chart

- Find which model has the highest classification accuracy

Confusion Matrix

- Show the confusion matrix of the best performing model with an explanation

Conclusions

- Point 1
- Point 2
- Point 3
- Point 4
- …

Appendix

- Include any relevant assets like Python code snippets, SQL queries, charts, Notebook outputs, or data sets that you may have created during this project

Thank you!